

KARLOVY VARY, Czech Republic

WMO: 114140

Lat: 50.2017N

Lon: 12.9142E

Elev: 607

StdP: 94.24

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.8	-16.2	1.0	-12.1	-13.4	1.3	-9.3	9.6	2.8	8.9	2.7	1.7	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.6	18.3	26.6	17.5	24.8	16.7	19.3	26.0	18.4	24.6	17.5	23.1	2.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.1	21.4	16.2	12.4	20.4	15.4	11.8	19.6	57.5	26.0	54.4	24.8	51.6	23.0	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.1	6.2	-16.3	32.0	2.5	1.8	-18.1	33.3	-19.6	34.3	-21.1	35.3	-22.9	36.5		
(5)			-16.6	21.2	2.5	1.3	-18.3	22.1	-19.8	22.9	-21.1	23.6	-22.9	24.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-2.1	-1.0	2.6	7.4	11.6	14.9	16.9	16.6	12.2	7.6	2.9	-1.0	(6)	(7)
	DBStd	7.82	4.60	4.65	4.06	4.24	3.78	3.76	3.46	3.33	3.49	3.86	3.47	4.32	(7)	(8)
	HDD10.0	1708	376	307	231	98	26	4	0	1	17	93	215	339	(8)	(9)
	HDD18.3	4041	634	541	489	328	209	114	71	74	186	333	464	598	(9)	(10)
	CDD10.0	768	0	0	1	21	77	151	213	206	82	19	1	0	(10)	(11)
	CDD18.3	61	0	0	0	0	2	12	25	21	1	0	0	0	(11)	(12)
	CDH23.3	728	0	0	0	2	29	152	288	233	24	0	0	0	(12)	(13)
	CDH26.7	170	0	0	0	0	3	31	77	58	1	0	0	0	(13)	(14)
(14) Wind	WSAvg	2.7	3.0	3.0	3.1	2.7	2.7	2.5	2.5	2.2	2.3	2.4	2.6	2.9	(14)	(15)
(15) Precipitation	PrecAvg	692	51	43	48	39	60	75	83	71	60	51	56	56	(15)	(16)
	PrecMax	920	119	101	136	85	134	123	141	165	134	160	111	120	(16)	(17)
	PrecMin	495	6	7	8	11	22	31	25	28	26	14	0	26	(17)	(18)
	PrecStd	97	26	25	30	19	30	24	28	31	27	28	30	27	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	11.2	16.3	22.7	26.4	30.0	31.6	31.3	26.1	20.2	13.1	9.4	(19)	(20)
		MCWB	7.1	6.6	8.9	13.0	16.7	19.2	18.4	18.7	16.7	14.3	10.6	7.8	(20)	(21)
	2%	DB	7.5	8.7	13.6	19.9	23.6	27.2	28.7	28.1	23.3	17.2	11.0	7.9	(21)	(22)
		MCWB	6.1	6.1	7.8	11.6	15.3	17.8	18.0	18.3	16.0	13.0	9.0	6.7	(22)	(23)
	5%	DB	5.6	7.0	11.0	17.6	21.1	25.0	26.7	25.8	21.0	15.1	9.4	6.2	(23)	(24)
		MCWB	4.4	5.0	6.7	10.7	14.0	16.8	17.4	17.3	15.0	12.1	8.0	5.0	(24)	(25)
	10%	DB	4.0	5.4	8.7	15.3	19.0	22.7	24.3	23.8	18.6	13.2	7.9	4.5	(25)	(26)
		MCWB	3.1	3.8	6.0	9.5	12.8	15.8	16.7	16.5	13.9	11.1	6.7	3.7	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	7.8	10.1	14.1	18.0	20.7	20.2	20.4	18.1	15.0	11.0	8.2	(27)	(28)
		MCDB	8.7	9.6	14.4	20.3	24.7	27.5	28.1	27.9	23.3	18.1	12.8	9.0	(28)	(29)
	2%	WB	6.2	6.5	8.7	12.5	16.1	18.9	19.1	19.1	16.6	13.6	9.5	6.8	(29)	(30)
		MCDB	7.3	8.2	11.9	18.4	21.5	25.0	25.8	25.9	21.7	16.4	10.6	7.8	(30)	(31)
	5%	WB	4.6	5.2	7.5	11.2	14.8	17.5	18.2	18.1	15.5	12.5	8.2	5.1	(31)	(32)
		MCDB	5.6	6.8	10.2	16.7	19.9	23.2	24.5	24.1	20.0	14.6	9.2	6.0	(32)	(33)
	10%	WB	3.1	3.8	6.2	10.0	13.6	16.4	17.3	17.2	14.5	11.3	6.8	3.8	(33)	(34)
		MCDB	3.9	5.3	8.6	14.5	18.1	21.7	22.9	22.4	18.0	13.1	7.8	4.5	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	6.0	7.6	9.9	10.1	10.5	10.4	10.7	9.3	7.4	4.8	4.3	(35)	(36)
		MCDBR	5.8	8.2	12.7	14.2	13.9	14.7	14.7	14.6	13.5	10.5	6.9	5.8	(36)	(37)
	5% WB	MCWBR	4.7	5.7	7.4	7.3	6.9	6.8	5.9	6.2	6.6	6.5	5.2	4.8	(37)	(38)
		MCDWR	5.4	6.9	10.8	13.1	12.5	13.4	12.6	12.8	11.9	9.5	6.2	5.4	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.284	0.306	0.344	0.382	0.384	0.385	0.397	0.390	0.361	0.336	0.310	0.281	(39)	(40)
		taud	2.417	2.386	2.352	2.293	2.313	2.330	2.298	2.336	2.399	2.460	2.482	2.442	(40)	(41)
	Ebn at Noon		766	836	859	860	872	873	855	842	831	790	729	716	(41)	(42)
		Edh at Noon	61	81	101	119	122	121	123	113	96	75	58	52	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		0.83	1.65	2.62	4.08	4.85	5.28	5.03	4.40	3.14	1.87	0.90	0.64	(43)	(44)
	RadStd		0.15	0.24	0.37	0.53	0.58	0.46	0.54	0.43	0.44	0.24	0.11	0.08	(44)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.40	N/A	N/A	N/A	+0.33	+0.35	N/A	N/A	+59	+22

Nomenclature: See separate page